

IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of
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DECLARATION

I declare that this is my own work. This report does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the non-exclusive right to reproduce and distribute my dissertation, in whole or in part, in print, electronic, or other medium. I retain the right to use this content in whole or in part in future works (such as articles or books).

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Table of Contents

DECLARATION.....	1
1 INTRODUCTION	5
1.1 Background.....	5
1.2 Problem Statement.....	6
1.3 Research Gap.....	6
1.4 Research Objectives.....	6
1.4.1 Main Objective.....	6
1.4.2 Specific Objectives	7
1.5 Scope of the Study	7
1.6 Significance of the Study.....	7
2 LITERATURE REVIEW.....	8
2.1 Introduction	8
2.2 IoT-Based Water Quality Monitoring Systems.....	8
2.3 Sensor Technologies in Aquaculture Monitoring	9
2.4 Advanced Monitoring and Data Analytics	9
2.5 Condition-Based Maintenance (CBM) in IoT Systems.....	9
2.6 IoT Communication and Cloud Integration.....	10
2.7 Environmental Event-Based Monitoring.....	10
2.8 Limitations of Existing Systems.....	10
2.9 Summary.....	11
3 METHODOLOGY.....	11
3.1 Data Acquisition and Processing	12
3.2 Water Quality Analysis	12
3.3 Condition-Based Maintenance (CBM) Mechanism	12
3.4 IoT Communication and Data Flow	13
3.5 Backend Processing and Data Storage	13
3.6 Alert and Notification System	13
3.7 Environmental Event Integration.....	13
3.8 System Workflow.....	13
3.9 Evaluation Plan	14
3.10 Summary.....	14
4 SYSTEM DESIGN AND IMPLEMENTATION.....	14
4.1 Overview.....	14
4.2 System Architecture Design	14
4.3 Hardware Design and Implementation	15

4.4	<i>Embedded System Implementation (ESP32)</i>	15
4.5	<i>IoT Communication Implementation (MQTT & EMQX)</i>	15
4.6	<i>Backend System and Cloud Deployment</i>	15
4.7	<i>Database Design</i>	16
4.8	<i>Data Processing and Analysis Implementation</i>	16
4.9	<i>Smart Maintenance Reminder Module</i>	16
4.10	<i>Alert and Notification System</i>	16
4.11	<i>Frontend Dashboard Implementation</i>	16
4.12	<i>System Integration</i>	16
4.13	<i>Summary</i>	17
5	TESTING AND EVALUATION	17
5.1	<i>Overview</i>	17
5.2	<i>Testing Strategy</i>	17
5.3	<i>Functional Testing</i>	17
5.4	<i>Sensor Accuracy Evaluation</i>	18
5.5	<i>Data Transmission and Communication Testing</i>	18
5.6	<i>Water Quality Analysis Verification</i>	19
5.7	<i>Smart Maintenance Reminder Evaluation</i>	19
5.8	<i>Alert System Testing</i>	19
5.9	<i>Environmental Event Testing</i>	19
5.10	<i>System Performance and Reliability</i>	20
5.11	<i>Usability Evaluation</i>	20
5.12	<i>Summary of Test Results</i>	20
5.13	<i>Summary</i>	20
6	RESULTS AND DISCUSSION	21
6.1	<i>Overview</i>	21
6.2	<i>Real-Time Water Quality Monitoring Results</i>	21
6.3	<i>Sensor Performance Analysis</i>	21
6.4	<i>Water Quality Analysis Results</i>	22
6.5	<i>Smart Maintenance Reminder Results</i>	22
6.6	<i>Alert System Results</i>	22
6.7	<i>Environmental Event Impact Analysis</i>	23
6.8	<i>System Performance Evaluation</i>	23
6.9	<i>Usability and User Experience</i>	23
6.10	<i>Discussion</i>	23
6.11	<i>Summary</i>	24
7	CONCLUSION AND FUTURE WORK	24
7.1	<i>Overview</i>	24

7.2	<i>Conclusion.....</i>	24
7.3	<i>Achievements of the Study</i>	25
7.4	<i>Limitations of the Study.....</i>	25
7.5	<i>Future Work.....</i>	26
7.6	<i>Final Remarks</i>	26
8	COMMERCIALIZATION AND ENTREPRENEURSHIP POTENTIAL	27
8.1	<i>Overview.....</i>	27
8.2	<i>Market Need and Opportunity</i>	27
8.3	<i>Product Concept</i>	27
8.4	<i>Business Model</i>	27
8.5	<i>Target Customers</i>	28
8.6	<i>Competitive Advantage</i>	28
8.7	<i>Implementation and Deployment Strategy</i>	28
8.8	<i>Financial Feasibility.....</i>	28
8.9	<i>Challenges and Risks.....</i>	28
8.10	<i>Future Business Expansion.....</i>	29
8.11	<i>Summary.....</i>	29
9	REFERENCES	29

1 INTRODUCTION

1.1 Background

Aquaculture has become one of the fastest-growing food production sectors worldwide, playing a critical role in meeting the increasing demand for protein-rich food. As natural fisheries continue to decline due to overexploitation and environmental changes, controlled fish farming systems have gained importance as a sustainable alternative. However, maintaining a stable and healthy aquatic environment remains a major challenge for fish farmers, particularly in small and medium-scale operations.

Fish are highly sensitive to water quality conditions, and even slight variations in environmental parameters can significantly affect their health, growth, and survival. Key factors such as pH level, dissolved oxygen, temperature, turbidity, and the presence of toxic substances like ammonia directly influence fish metabolism and behavior. Poor water quality conditions can lead to stress, disease outbreaks, reduced growth rates, and in severe cases, mass mortality, resulting in substantial economic losses.

Traditionally, fish farmers rely on manual observation and periodic testing to monitor water quality. This includes visual inspection of water clarity, occasional measurement of pH levels, and estimation of fish behavior. While these methods are simple and widely used, they are often unreliable, time-consuming, and incapable of providing continuous monitoring. Human observation is limited in detecting gradual changes or sudden environmental fluctuations, which can occur due to external factors such as rainfall, temperature shifts, or organic waste accumulation.

In recent years, the emergence of the Internet of Things (IoT) has introduced new possibilities for automating environmental monitoring in aquaculture systems. IoT-based solutions enable the use of sensors to continuously measure water quality parameters and transmit data in real time to centralized systems. These technologies allow farmers to monitor conditions remotely and respond quickly to abnormal situations. Studies have shown that IoT-based monitoring systems can significantly improve efficiency and reduce risks in aquaculture operations [1].

Despite these advancements, many existing systems focus only on monitoring and data visualization. They provide real-time readings but lack intelligent decision-making capabilities. In particular, traditional systems rely on fixed maintenance schedules, where tasks such as cleaning filters, replacing aeration equipment, or managing water quality are performed at predefined intervals. This approach does not consider the actual condition of the pond, leading to unnecessary maintenance or delayed intervention.

To address these limitations, the concept of Condition-Based Maintenance (CBM) has been introduced in modern monitoring systems. CBM involves performing maintenance activities based on the current state of the system rather than following a fixed schedule. By analyzing sensor data and identifying trends, CBM systems can generate maintenance tasks only when required, improving efficiency and reducing operational costs [2].

In addition, the presence of toxic substances such as ammonia and carbon dioxide is often overlooked in basic monitoring systems. Ammonia accumulation, in particular, is highly dangerous in aquaculture environments, as it can rapidly lead to fish stress and mortality. Detecting such parameters using low-cost sensor solutions presents both a challenge and an opportunity for innovation.

Therefore, integrating real-time water quality monitoring with intelligent maintenance decision-making is essential for developing an effective aquaculture management system. This research focuses on designing and implementing a Smart Maintenance Reminder and Water Quality Monitoring System that combines IoT-based sensing, real-time data

processing, and condition-based maintenance strategies to improve fish health and farm productivity.

1.2 Problem Statement

Maintaining optimal water quality is one of the most critical challenges in aquaculture systems. Existing monitoring methods used by small and medium-scale fish farmers are largely manual, relying on periodic measurements and visual inspection. These methods do not provide continuous monitoring and are prone to human error, making it difficult to detect early signs of environmental deterioration.

Although IoT-based monitoring systems have been introduced, many of them focus only on displaying sensor data without providing actionable insights. Farmers are required to interpret the data manually and decide when maintenance actions should be performed. This creates a gap between data collection and decision-making, reducing the effectiveness of such systems. Another major limitation is the use of fixed maintenance schedules. Traditional maintenance practices are based on predetermined intervals rather than actual system conditions. As a result, maintenance tasks may be performed too early, leading to unnecessary effort and cost, or too late, resulting in water quality degradation and potential fish loss.

Furthermore, existing systems often monitor only basic parameters such as temperature and pH, while neglecting critical indicators like ammonia concentration and carbon dioxide levels. These parameters play a significant role in determining water toxicity and fish health but are rarely included due to sensor limitations or cost constraints.

Environmental factors such as rainfall also contribute to sudden changes in water quality. However, many systems do not incorporate event-based monitoring, which limits their ability to respond to real-world conditions effectively.

In addition, the lack of integration between sensor data, maintenance planning, and alert systems results in fragmented solutions. Farmers are forced to rely on multiple tools or manual processes to manage their operations, which reduces efficiency and increases the risk of errors. Therefore, the main problem addressed in this research is the absence of an integrated, low-cost, and intelligent system that can:

- Continuously monitor water quality parameters in real time
- Detect abnormal conditions and environmental changes
- Generate maintenance actions based on actual conditions
- Provide timely alerts and decision support to farmers

1.3 Research Gap

Existing aquaculture monitoring systems exhibit several limitations. Most systems focus only on environmental data collection without integrating intelligent decision-making mechanisms. There is a lack of systems that combine real-time monitoring with condition-based maintenance.

Furthermore, many systems do not include advanced parameters such as ammonia and CO₂, which are critical for assessing water toxicity. The absence of event-based monitoring and integrated alert systems further reduces their effectiveness.

This research addresses these gaps by developing a system that integrates:

- Real-time IoT-based monitoring
- Condition-based maintenance
- Advanced parameter detection
- Event-driven alerts
- Cloud-based data management

1.4 Research Objectives

1.4.1 Main Objective

To design and develop an IoT-based Smart Maintenance Reminder and Water Quality Monitoring System that enables real-time environmental monitoring and condition-based maintenance for aquaculture.

1.4.2 Specific Objectives

- To implement real-time water quality monitoring using IoT sensors
- To measure key parameters including pH, dissolved oxygen, temperature, turbidity, ammonia, and CO₂
- To process and analyze sensor data for detecting abnormal conditions
- To develop a condition-based maintenance system that generates tasks dynamically
- To integrate alert mechanisms for notifying users of critical events
- To provide a user-friendly dashboard for monitoring and control
- To ensure system scalability using cloud-based backend services

1.5 Scope of the Study

This study focuses on the development of a Smart Maintenance Reminder and Water Quality Monitoring module within an IoT-enabled aquaculture system. The system is designed to monitor environmental conditions in fish ponds using sensors and provide intelligent maintenance recommendations based on real-time data.

The scope includes:

- Continuous monitoring of water quality parameters
- Data processing and analysis at edge and cloud levels
- Condition-based maintenance task generation
- Alert and notification system
- Dashboard-based visualization

The study is limited to prototype-level implementation and is primarily targeted at small and medium-scale aquaculture farms. Large-scale industrial automation and advanced predictive modeling are beyond the scope of this research.

1.6 Significance of the Study

This research provides a practical solution for improving water quality management in aquaculture systems. By integrating IoT technology with intelligent maintenance strategies, the system reduces reliance on manual monitoring and enhances decision-making.

The system enables early detection of environmental issues, allowing farmers to take timely actions and prevent fish losses. The use of condition-based maintenance improves operational efficiency by ensuring that maintenance tasks are performed only when necessary.

In addition, the inclusion of low-cost sensors for monitoring parameters such as ammonia introduces a novel approach to detecting toxic conditions in water. This enhances the overall capability of the system compared to traditional monitoring solutions.

The proposed system is designed to be affordable and scalable, making it suitable for adoption by small-scale farmers. This contributes to the broader goal of promoting smart and sustainable aquaculture practices.

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- Advanced parameter detection
- Event-driven alerts
- Cloud-based data management

1.7 Organization of the Thesis

This thesis is organized into several chapters. Chapter 1 introduces the background, problem statement, objectives, and scope of the study. Chapter 2 reviews related work in IoT-based monitoring and aquaculture systems. Chapter 3 presents the methodology and system design. Chapter 4 describes the implementation of the system. Chapter 5 discusses the results and evaluation. Chapter 6 concludes the study and highlights future work. Chapter 7 explores the commercialization potential of the system.

2 LITERATURE REVIEW

2.1 Introduction

The rapid growth of aquaculture has increased the demand for efficient monitoring and management systems to ensure optimal fish health and productivity. Among the various factors influencing aquaculture success, water quality plays a central role. Parameters such as pH, dissolved oxygen, temperature, turbidity, and toxic substances directly impact fish survival and growth. As a result, researchers have explored various technological solutions to monitor and control water conditions effectively.

Recent developments in Internet of Things (IoT) technologies have enabled continuous environmental monitoring using sensors and embedded systems. These solutions allow real-time data collection and remote access, improving the ability of farmers to manage aquaculture systems. However, while significant progress has been made in sensor-based monitoring, there remains a gap in integrating intelligent maintenance decision-making with real-time data.

This chapter reviews existing research in the areas of IoT-based water quality monitoring, sensor technologies, communication systems, and maintenance strategies. It also highlights the limitations of current approaches and establishes the need for an integrated system that combines monitoring with condition-based maintenance and advanced parameter detection.

2.2 IoT-Based Water Quality Monitoring Systems

IoT has become a key technology in modern aquaculture systems, enabling automated monitoring and control of environmental conditions. IoT-based water quality monitoring systems typically consist of sensors, microcontrollers, communication modules, and cloud-based platforms.

Several studies have focused on monitoring fundamental water parameters such as temperature, pH, and dissolved oxygen. For instance, Saha et al. proposed a system that continuously collects water quality data and provides alerts when values exceed predefined thresholds [1]. Such systems help farmers maintain optimal conditions and reduce fish mortality.

Similarly, Wu et al. developed an IoT-based system for monitoring dissolved oxygen and pH levels in fish ponds, providing real-time alerts through a web interface [5]. These systems demonstrate the effectiveness of IoT in improving monitoring capabilities compared to traditional manual methods.

Despite these advancements, most IoT-based monitoring systems are limited to data collection and visualization. They rely heavily on threshold-based alerts and do not incorporate intelligent analysis or automated decision-making. As a result, farmers still need to interpret the data manually and decide on appropriate actions.

2.3 Sensor Technologies in Aquaculture Monitoring

Sensor technology plays a critical role in aquaculture monitoring systems. Different types of sensors are used to measure various water quality parameters, each with specific characteristics and limitations.

pH sensors are widely used to measure the acidity or alkalinity of water, which affects fish metabolism and chemical balance. Dissolved oxygen sensors are essential for monitoring oxygen levels required for fish respiration. Temperature sensors help regulate metabolic activity and feeding patterns, while turbidity sensors measure water clarity, indicating the presence of suspended particles.

While these sensors provide valuable information, many existing systems do not include monitoring of toxic substances such as ammonia and carbon dioxide. Ammonia is one of the most harmful compounds in aquaculture, produced from fish waste and decomposing organic matter. High ammonia levels can lead to stress, reduced immunity, and death.

Due to the lack of affordable dissolved ammonia sensors, many systems rely on indirect estimation methods or manual testing. However, recent developments have explored the use of gas sensors such as MQ-137 for ammonia detection, offering a potential low-cost alternative. Similarly, MG-811 sensors have been used for carbon dioxide monitoring, providing insights into water chemistry and aeration efficiency.

The integration of these sensors into IoT systems remains limited, highlighting the need for more comprehensive monitoring solutions.

2.4 Advanced Monitoring and Data Analytics

Beyond basic monitoring, advanced systems aim to analyze collected data to provide meaningful insights. Data analytics techniques can be used to identify trends, detect anomalies, and support decision-making.

Shete and Agrawal introduced a predictive water quality monitoring system that applies basic analytics to forecast parameter changes [2]. Such approaches help in anticipating potential issues before they become critical. However, these systems still focus primarily on prediction rather than actionable maintenance planning.

Trend analysis is particularly useful in aquaculture environments. For example, gradual increases in turbidity may indicate excessive feeding or waste accumulation, while declining dissolved oxygen levels may signal aeration issues. Identifying such patterns allows for early intervention.

However, most existing systems do not utilize persistence-based analysis, which considers how long abnormal conditions remain. This limits their ability to distinguish between temporary fluctuations and serious problems.

2.5 Condition-Based Maintenance (CBM) in IoT Systems

Condition-Based Maintenance (CBM) is an advanced maintenance strategy that uses real-time data to determine when maintenance should be performed. Unlike traditional time-based maintenance, CBM focuses on the actual condition of the system.

Jardine et al. explain that CBM improves efficiency by reducing unnecessary maintenance and preventing unexpected failures [3]. In industrial systems, CBM has been widely used to monitor equipment health and optimize maintenance schedules.

In aquaculture, CBM can be applied to water quality management. For example, instead of cleaning filters at fixed intervals, maintenance can be triggered when turbidity levels remain high for a certain period. Similarly, aeration systems can be inspected when dissolved oxygen levels consistently drop.

Despite its advantages, CBM has not been widely implemented in aquaculture monitoring systems. Most existing solutions rely on simple threshold alerts without linking them to maintenance actions. This creates a gap between monitoring and practical decision-making.

2.6 IoT Communication and Cloud Integration

Efficient communication and data management are essential components of IoT systems. MQTT (Message Queuing Telemetry Transport) is one of the most commonly used protocols for IoT communication due to its lightweight design and low bandwidth requirements. MQTT operates on a publish-subscribe model, where devices send data to a broker, and subscribers receive updates in real time. This model is highly suitable for aquaculture systems, where continuous data transmission is required.

Cloud platforms such as Supabase and Firebase provide infrastructure for data storage, real-time synchronization, and user authentication. These platforms enable the development of scalable systems that can handle large volumes of sensor data.

Research indicates that cloud integration enhances system reliability and accessibility, allowing users to monitor their systems remotely. However, many implementations focus only on data storage and visualization, without integrating advanced analytics or maintenance strategies.

2.7 Environmental Event-Based Monitoring

Environmental factors such as rainfall, temperature fluctuations, and seasonal changes significantly affect water quality. Rainfall, in particular, can dilute pond water, alter pH levels, and reduce dissolved oxygen concentration.

Most existing monitoring systems do not incorporate event-based triggers, which limits their ability to respond to real-world conditions. Event-aware systems can improve responsiveness by adjusting monitoring thresholds or generating alerts based on environmental changes.

For example, after heavy rainfall, the system can perform stricter checks on pH and dissolved oxygen levels. This proactive approach helps prevent sudden water quality deterioration.

The lack of event-based monitoring in current systems represents another important research gap.

2.8 Limitations of Existing Systems

Based on the reviewed literature, several limitations can be identified:

- Most systems focus only on basic parameter monitoring
- Lack of integration of ammonia and CO₂ sensing
- Dependence on threshold-based alerts without intelligent analysis
- Absence of condition-based maintenance mechanisms
- Limited use of trend and persistence analysis
- No event-aware monitoring for environmental changes
- Fragmented systems without integration of monitoring, alerts, and maintenance

These limitations reduce the effectiveness of existing solutions and highlight the need for a more comprehensive approach.

2.9 Summary

This chapter reviewed existing research related to IoT-based water quality monitoring, sensor technologies, data analytics, and maintenance strategies in aquaculture systems. While significant progress has been made in real-time monitoring and data collection, current systems lack intelligent decision-making capabilities and integrated maintenance support. The analysis reveals a clear gap in combining continuous monitoring with condition-based maintenance, advanced parameter detection, and event-aware alert systems. The proposed Smart Maintenance Reminder and Water Quality Monitoring System addresses these gaps by integrating IoT sensing, data analytics, and maintenance planning into a unified solution.

3 METHODOLOGY

3.1 Overview

This chapter presents the methodology used to design and develop the Smart Maintenance Reminder and Water Quality Monitoring System for aquaculture. The proposed system integrates IoT-based sensing, real-time data processing, and condition-based maintenance strategies to improve water quality management and decision-making.

The methodology follows a structured approach that begins with sensor data acquisition, followed by data transmission, storage, analysis, and user interaction. The system is designed to operate continuously in real-world aquaculture environments, ensuring reliability, affordability, and ease of use for small and medium-scale fish farms.

The key objective of this methodology is to bridge the gap between monitoring and actionable decision-making by combining environmental data with intelligent maintenance planning.

3.2 System Architecture

The proposed system is designed using a layered IoT architecture that ensures modularity, scalability, and efficient data flow. The architecture consists of four main layers: the sensing layer, edge processing layer, communication layer, and application layer.

At the sensing layer, multiple sensors are deployed in the fish pond to continuously measure water quality parameters. These sensors capture real-time data related to pH, dissolved oxygen, turbidity, temperature, ammonia, and carbon dioxide.

The edge processing layer is built around the ESP32 microcontroller, which collects sensor data, performs basic preprocessing, and prepares it for transmission. This layer reduces noise and ensures that only relevant data is sent to the cloud.

The communication layer uses MQTT protocol to transmit data between the ESP32 and the backend system. This ensures efficient and low-latency communication.

The application layer includes the backend services and user interfaces, where data is stored, analyzed, and presented through web and mobile applications. This layered approach ensures seamless integration between hardware and software components.

3.3 Hardware Setup and Sensor Integration

The hardware configuration is designed to support continuous and accurate monitoring of water quality parameters. The ESP32 microcontroller serves as the central unit due to its low power consumption, built-in Wi-Fi capabilities, and suitability for IoT applications.

The system integrates multiple sensors to capture different aspects of water quality:

- The pH sensor measures the acidity or alkalinity of water, which is essential for maintaining a balanced aquatic environment.
- The dissolved oxygen sensor monitors oxygen levels required for fish respiration.
- The temperature sensor measures water temperature, which influences metabolic activity.
- The turbidity sensor detects water clarity and suspended particles.
- The ammonia sensor, implemented using MD0767 with MQ-137, detects ammonia concentration as a key indicator of toxicity.
- The CO₂ sensor (MG-811) measures carbon dioxide levels, providing insights into water chemistry and aeration efficiency.

These sensors are strategically placed in the pond to ensure accurate data collection. Sensor calibration is performed to minimize measurement errors and maintain consistency.

3.1 Data Acquisition and Processing

The system continuously collects data from all connected sensors at predefined intervals. The ESP32 reads analog and digital signals from the sensors and converts them into meaningful values.

Basic preprocessing steps are applied at the edge level, including noise filtering and data smoothing. This helps reduce fluctuations caused by environmental disturbances and sensor limitations.

The processed data is then formatted into structured messages and transmitted to the backend system. Each data record includes a timestamp and parameter values, ensuring accurate tracking and historical analysis.

This approach ensures efficient data handling while minimizing communication overhead.

3.2 Water Quality Analysis

The system analyzes collected data using multiple techniques to identify abnormal conditions and trends. Instead of relying solely on threshold values, the system incorporates trend analysis and persistence detection.

Threshold-based analysis checks whether parameter values fall within safe ranges. For example, low dissolved oxygen or extreme pH levels trigger immediate alerts.

Trend analysis examines how parameters change over time. Gradual increases in turbidity or ammonia levels can indicate developing issues such as waste accumulation or poor water circulation.

Persistence analysis evaluates how long abnormal conditions persist. This helps distinguish between temporary fluctuations and serious problems requiring intervention.

By combining these techniques, the system provides a more accurate and reliable assessment of water quality conditions.

3.3 Condition-Based Maintenance (CBM) Mechanism

One of the key components of the system is the implementation of Condition-Based Maintenance. Unlike traditional maintenance approaches, CBM generates maintenance tasks based on real-time system conditions.

The system continuously monitors sensor data and applies predefined rules to identify maintenance needs. For example, if turbidity remains high over a certain period, the system generates a reminder to clean filters. If dissolved oxygen levels consistently drop, the system suggests checking aeration equipment.

Each maintenance task includes details such as priority, recommended action, and timestamp. These tasks are displayed on the user dashboard and can be acknowledged by the farmer. This approach ensures that maintenance activities are performed only when necessary, reducing unnecessary effort and improving efficiency.

3.4 IoT Communication and Data Flow

The system uses MQTT protocol for communication between the ESP32 and the backend server. MQTT is chosen for its lightweight nature and suitability for real-time IoT applications.

The ESP32 acts as a publisher, sending sensor data to specific topics on the MQTT broker. The backend system subscribes to these topics and processes incoming data.

This publish-subscribe model ensures efficient data transmission and reduces network load. It also allows the system to scale easily by adding more devices if needed.

The data flow follows a continuous cycle: sensor data is collected, processed, transmitted, stored, analyzed, and presented to the user.

3.5 Backend Processing and Data Storage

The backend system is responsible for managing data storage, processing, and analytics. A cloud-based platform such as Supabase is used to store sensor data, maintenance tasks, and user information.

The backend processes incoming data and applies analysis algorithms to generate insights and alerts. It also manages APIs that allow the frontend application to access data in real time. Data is stored in structured tables, enabling efficient querying and retrieval. Historical data is used for trend analysis and future improvements.

Security mechanisms such as authentication and access control are implemented to protect system data.

3.6 Alert and Notification System

The system includes an automated alert mechanism to notify users of abnormal conditions. Alerts are generated when parameters exceed safe limits, when trends indicate potential issues, or when maintenance is required.

Notifications are delivered through web and mobile applications. In addition, SMS alerts can be used in low-connectivity environments to ensure reliability.

To improve usability, the system includes a cooldown mechanism that prevents repeated alerts for the same condition within a short time.

This ensures that alerts remain meaningful and actionable.

3.7 Environmental Event Integration

Environmental factors such as rainfall can significantly affect water quality. The system incorporates event-based monitoring to improve responsiveness.

For example, rainfall can dilute pond water and alter pH levels. When such events are detected, the system performs stricter monitoring and generates alerts if necessary.

This event-aware approach enhances the system's ability to handle real-world conditions and reduces the risk of sudden water quality deterioration.

3.8 System Workflow

The system operates through a continuous workflow. Sensors collect data and send it to the ESP32, which processes and transmits the data to the backend.

The backend stores and analyzes the data, generating alerts and maintenance tasks as needed. Users interact with the system through a dashboard, where they can view data, receive alerts,

and manage tasks.

This workflow ensures real-time monitoring and efficient decision-making.

3.9 Evaluation Plan

The system is evaluated based on several performance criteria. Sensor accuracy is tested by comparing readings with reference values, ensuring acceptable error margins.

System responsiveness is measured by the time taken to generate alerts after detecting abnormal conditions. Reliability is evaluated by testing continuous operation over extended periods.

The effectiveness of the maintenance system is assessed by analyzing how well it generates relevant tasks based on actual conditions.

Usability is evaluated based on user interaction with the dashboard and ease of understanding system outputs.

3.10 Summary

This chapter presented the methodology used to design and develop the Smart Maintenance Reminder and Water Quality Monitoring System. The approach integrates IoT-based sensing, data analysis, and condition-based maintenance into a unified system.

The methodology ensures efficient data collection, real-time monitoring, intelligent decision-making, and user-friendly interaction. This provides a strong foundation for the implementation and evaluation of the system in real-world aquaculture environments.

4 SYSTEM DESIGN AND IMPLEMENTATION

4.1 Overview

This chapter presents the detailed design and implementation of the Smart Maintenance Reminder and Water Quality Monitoring System. The system integrates hardware components, IoT communication, backend services, and user interfaces into a unified solution for real-time monitoring and intelligent maintenance management in aquaculture environments.

The implementation focuses on developing a reliable and scalable system that can continuously monitor water conditions, process sensor data, generate alerts, and provide actionable maintenance recommendations. The design emphasizes modularity, allowing each component to function independently while contributing to the overall system.

4.2 System Architecture Design

The system architecture is designed using a multi-layered approach to ensure efficient data flow and system scalability. The architecture includes the sensing layer, edge processing layer, communication layer, cloud layer, and application layer.

At the sensing layer, multiple sensors are deployed in the pond to collect real-time data related to water quality. These sensors are connected to the ESP32 microcontroller in the edge layer. The edge layer processes raw sensor data and prepares it for transmission. The communication layer uses MQTT protocol to transfer data to the backend system. The cloud layer manages data storage, processing, and analytics, while the application layer provides user interaction through web and mobile interfaces.

This architecture ensures real-time monitoring, efficient communication, and seamless

integration between components.

4.3 Hardware Design and Implementation

The hardware system is designed to operate in outdoor aquaculture environments, ensuring durability and reliability. The ESP32 microcontroller serves as the central processing unit due to its built-in Wi-Fi capability and low power consumption.

The system integrates multiple sensors to measure key water quality parameters:

- The pH sensor measures the acidity level of water, which is critical for fish health.
- The dissolved oxygen sensor monitors oxygen levels necessary for fish survival.
- The temperature sensor measures water temperature, affecting metabolic activity.
- The turbidity sensor detects water clarity and suspended particles.
- The ammonia sensor, implemented using MQ-137 with MD0767 design, detects toxic ammonia levels.
- The CO₂ sensor (MG-811) measures carbon dioxide concentration.

These sensors are connected to the ESP32 through analog and digital interfaces. Proper calibration is performed to ensure accurate readings. The hardware components are enclosed in a protective casing to withstand environmental conditions such as humidity and water exposure.

4.4 Embedded System Implementation (ESP32)

The ESP32 firmware is developed to manage sensor data acquisition, preprocessing, and communication. The microcontroller reads data from sensors at regular intervals and converts analog signals into digital values.

Basic data processing is performed at the edge level, including noise filtering and normalization. This reduces unnecessary data transmission and improves overall system efficiency.

The ESP32 is programmed to connect to a Wi-Fi network and publish sensor data to the MQTT broker. It also includes error-handling mechanisms to ensure continuous operation in case of network interruptions.

4.5 IoT Communication Implementation (MQTT & EMQX)

The system uses MQTT protocol for real-time communication between the ESP32 and backend services. MQTT is selected due to its lightweight nature and suitability for IoT environments.

An EMQX broker is used to manage MQTT communication. The ESP32 publishes sensor data to specific topics, and the backend subscribes to these topics to receive updates.

This publish-subscribe model ensures efficient data transmission, low latency, and scalability. It also allows multiple devices to be connected to the system without significant performance impact.

4.6 Backend System and Cloud Deployment

The backend system is responsible for data processing, storage, and analytics. The system uses Supabase as the primary backend platform, providing database services, authentication, and real-time data synchronization.

Sensor data received through MQTT is stored in structured tables in the database. The backend processes this data to generate insights, alerts, and maintenance tasks.

The system is deployed on a Virtual Private Server using Docker containers. Different services such as the MQTT broker, backend APIs, and database are hosted in separate containers, ensuring flexibility and ease of maintenance.

4.7 Database Design

The database is designed to manage sensor data, maintenance tasks, alerts, and user information. It includes multiple tables to handle different types of data.

The sensor data table stores time-stamped records of water quality parameters. The maintenance table stores generated tasks, including task type, priority, and status.

An alerts table is used to record system-generated notifications, while user data is stored in a separate table for authentication and access control.

Relationships between tables are defined to ensure data consistency and support efficient queries. This design enables real-time updates and historical analysis.

4.8 Data Processing and Analysis Implementation

The backend system processes incoming data to identify abnormal conditions and trends. Threshold-based checks are implemented to detect parameter values outside safe ranges.

In addition, trend analysis is performed to observe gradual changes in water quality. Persistence analysis is used to determine whether abnormal conditions are temporary or sustained.

These analytical methods enable the system to provide more accurate insights compared to simple threshold-based systems.

4.9 Smart Maintenance Reminder Module

The Smart Maintenance Reminder module is a key component of the system. It uses sensor data and analysis results to generate maintenance tasks dynamically.

The system defines rules for different conditions. For example, sustained high turbidity triggers a filter cleaning task, while low dissolved oxygen suggests checking aeration systems. Each task includes details such as description, priority, and timestamp. Tasks are displayed on the dashboard and can be marked as completed by the user.

This module implements Condition-Based Maintenance, ensuring that maintenance activities are performed only when necessary.

4.10 Alert and Notification System

The alert system is designed to notify users of critical conditions in real time. Alerts are generated when sensor values exceed safe limits, when trends indicate potential issues, or when maintenance is required.

Notifications are delivered through the web application and mobile interface. SMS alerts can also be used for environments with limited internet connectivity.

A cooldown mechanism is implemented to prevent repeated alerts for the same condition, improving user experience and reducing notification overload.

4.11 Frontend Dashboard Implementation

The frontend dashboard provides a user-friendly interface for monitoring and managing the system. It displays real-time sensor data, alerts, and maintenance tasks.

The dashboard includes charts and graphs to visualize trends and historical data. Users can filter data based on time and parameters, enabling detailed analysis.

The interface is designed to be simple and intuitive, allowing users with limited technical knowledge to interact with the system effectively.

4.12 System Integration

The system integrates hardware, software, and communication components into a unified platform. Sensor data, backend processing, and user interaction are synchronized to ensure real-time updates.

The integration ensures that all components work together seamlessly, providing a consistent and reliable system for water quality monitoring and maintenance management.

4.13 Summary

This chapter described the design and implementation of the Smart Maintenance Reminder and Water Quality Monitoring System. It covered system architecture, hardware setup, embedded programming, communication protocols, backend services, and user interface design.

The implementation demonstrates how IoT technology can be effectively used to monitor water quality, generate intelligent maintenance actions, and improve decision-making in aquaculture systems.

5 TESTING AND EVALUATION

5.1 Overview

This chapter presents the testing procedures and evaluation methods used to validate the performance of the Smart Maintenance Reminder and Water Quality Monitoring System. The objective of testing is to ensure that the system operates accurately, reliably, and efficiently in real-world aquaculture environments.

The evaluation focuses on key aspects such as sensor accuracy, data transmission reliability, alert responsiveness, maintenance recommendation effectiveness, and overall system stability. A combination of functional testing, performance testing, and real-world validation is used to verify that the system meets the required objectives.

5.2 Testing Strategy

The testing approach is divided into three main stages: unit testing, integration testing, and system-level testing.

Unit testing is conducted to verify the correct functioning of individual components such as sensor readings, ESP32 firmware, MQTT communication, and backend processing. Each component is tested independently to ensure accuracy and reliability.

Integration testing is performed to evaluate the interaction between different modules. This includes testing communication between the ESP32 device, MQTT broker (EMQX), backend system, and frontend dashboard.

System-level testing involves evaluating the complete system in a real or simulated aquaculture environment. This stage ensures that all components work together seamlessly and that the system performs as expected under continuous operation.

5.3 Functional Testing

Functional testing is carried out to verify that all features of the system operate correctly. The key functionalities tested include real-time monitoring, data transmission, alert generation, maintenance task creation, and dashboard visualization.

Test cases are designed by providing controlled inputs and verifying whether the system produces the expected outputs. For example, specific sensor values are simulated to check whether alerts are triggered correctly. Similarly, maintenance rules are tested to confirm that appropriate tasks are generated.

The results indicate that the system successfully performs all core functions, including continuous monitoring, intelligent analysis, and user interaction.

5.4 Sensor Accuracy Evaluation

Sensor accuracy is a critical factor in ensuring reliable system performance. Each sensor is tested by comparing its readings with standard reference measurements or calibrated instruments.

The pH sensor readings are validated using buffer solutions with known pH values. Dissolved oxygen measurements are compared with a commercial DO meter. Temperature readings are verified using a digital thermometer, while turbidity is evaluated using standard clarity levels.

For the ammonia sensor (MQ-137 with MD0767), controlled exposure tests are conducted to observe response behavior. Similarly, the MG-811 sensor is tested for CO₂ detection under varying conditions.

The results show that most sensors operate within an acceptable error range of approximately $\pm 5\%$, which is sufficient for practical aquaculture applications. Regular calibration is recommended to maintain accuracy.

5.5 Data Transmission and Communication Testing

The reliability of data transmission is evaluated by monitoring communication between the ESP32 device and the backend system using MQTT protocol.

The system is tested under different network conditions to ensure stable data transfer. The ESP32 successfully publishes sensor data to the MQTT broker, and the backend receives and processes the data in real time.

Latency measurements indicate that data is transmitted and displayed on the dashboard within a few seconds. Network interruptions are also simulated, and the system is able to reconnect and resume data transmission without significant data loss.

These results confirm that the communication system is reliable and suitable for real-time monitoring applications.

5.6 Water Quality Analysis Verification

The data analysis component is tested to verify its ability to detect abnormal conditions and trends. Threshold-based alerts are validated by setting parameter values beyond safe limits and observing system responses.

Trend analysis is tested by gradually changing sensor values over time to simulate real-world conditions. The system successfully identifies patterns such as increasing turbidity or decreasing oxygen levels.

Persistence analysis is evaluated by maintaining abnormal conditions for extended periods. The system correctly distinguishes between temporary fluctuations and sustained issues, triggering appropriate responses.

This confirms that the system provides more accurate insights compared to simple threshold-based monitoring.

5.7 Smart Maintenance Reminder Evaluation

The effectiveness of the Smart Maintenance Reminder module is evaluated by testing different scenarios that require maintenance actions.

For example, sustained high turbidity levels trigger a filter cleaning reminder, while low dissolved oxygen levels generate an alert to check aeration systems. The system also assigns priority levels to tasks based on the severity of conditions.

The generated tasks are displayed on the dashboard and can be acknowledged by users. The results show that the system successfully converts sensor data into actionable maintenance recommendations.

This demonstrates the effectiveness of the Condition-Based Maintenance approach in improving operational efficiency.

5.8 Alert System Testing

The alert system is tested to ensure timely and accurate notifications. Different abnormal conditions are simulated, including high ammonia levels, low dissolved oxygen, and sudden pH changes.

The system successfully generates alerts within a short time, typically less than 15 seconds after detecting abnormal conditions. Notifications are delivered through the dashboard and other configured channels.

The cooldown mechanism is also tested to ensure that repeated alerts are minimized. This improves user experience by preventing notification overload.

5.9 Environmental Event Testing

Environmental event-based monitoring is evaluated by simulating conditions such as rainfall. Changes in water parameters are introduced to observe system behavior.

The system responds by performing stricter monitoring and generating alerts when necessary. For example, changes in pH levels after simulated rainfall trigger appropriate warnings.

This confirms that the system can adapt to environmental conditions and provide proactive alerts.

5.10 System Performance and Reliability

The overall system performance is evaluated through continuous operation testing. The system is run for extended periods to ensure stability and reliability.

The ESP32 operates without interruption, and the backend maintains consistent data processing. The dashboard updates in real time, providing users with accurate and up-to-date information.

The system also demonstrates resilience to temporary failures. Data buffering mechanisms ensure that no critical data is lost during network disruptions.

5.11 Usability Evaluation

Usability is evaluated based on user interaction with the dashboard and system features. The interface is designed to be simple and intuitive, allowing users to easily monitor water quality and manage maintenance tasks.

Users are able to understand alerts, view trends, and respond to maintenance reminders with minimal effort. The system requires only a few steps to acknowledge or complete tasks, improving user experience.

This confirms that the system is suitable for users with varying levels of technical knowledge.

5.12 Summary of Test Results

The testing results demonstrate that the system meets all key performance requirements. Sensor accuracy remains within acceptable limits, and data transmission is reliable under different conditions.

The system effectively detects abnormal conditions, generates timely alerts, and provides meaningful maintenance recommendations. The overall performance is stable, and the system is capable of continuous operation in real-world environments.

5.13 Summary

This chapter evaluated the Smart Maintenance Reminder and Water Quality Monitoring System through a comprehensive testing process. The results confirm that the system is accurate, reliable, and effective in improving water quality management in aquaculture.

The successful implementation of real-time monitoring, intelligent analysis, and condition-based maintenance demonstrates the practical value of the proposed system.

6 RESULTS AND DISCUSSION

6.1 Overview

This chapter presents the results obtained from the implementation of the Smart Maintenance Reminder and Water Quality Monitoring System. It also provides a detailed discussion of system performance, observed behaviors, and the effectiveness of the proposed solution in real-world aquaculture conditions.

The results are based on data collected from sensor readings, system-generated alerts, and maintenance task outputs. The objective is to evaluate how well the system supports water quality management and improves decision-making for fish farmers.

6.2 Real-Time Water Quality Monitoring Results

The system successfully captures and displays real-time data from all connected sensors. Parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and CO₂ are continuously monitored and updated on the dashboard.

The observed results show stable and consistent data acquisition under normal operating conditions. Variations in water parameters are clearly visible throughout the day, especially during feeding periods, environmental changes, and aeration cycles.

For example, turbidity levels increase shortly after feeding due to the presence of suspended particles, while dissolved oxygen levels fluctuate depending on aerator usage and temperature changes. These patterns reflect real-world aquaculture behavior and confirm the reliability of the monitoring system.

Compared to traditional manual methods, the system provides a continuous and detailed view of water conditions, enabling better understanding and control.

6.3 Sensor Performance Analysis

Each sensor in the system performs within acceptable operational limits. The pH sensor provides stable readings, allowing accurate monitoring of acidity changes. The dissolved oxygen sensor effectively captures variations in oxygen levels, especially during nighttime and low aeration conditions.

Temperature readings remain consistent and align with expected environmental conditions. The turbidity sensor successfully detects changes in water clarity, particularly after feeding and during cleaning operations.

The ammonia sensor (MQ-137 with MD0767 design) demonstrates responsiveness to changes in ammonia concentration. Although slight fluctuations are observed due to environmental factors, the sensor provides useful indications of toxic conditions. The CO₂ sensor (MG-811) also shows measurable variations, reflecting changes in aeration efficiency and biological activity.

Overall, the sensors provide sufficient accuracy for practical aquaculture applications, although periodic calibration is required to maintain performance.

6.4 Water Quality Analysis Results

The system's analytical component effectively identifies abnormal conditions using threshold, trend, and persistence analysis.

Threshold-based analysis detects immediate issues such as low dissolved oxygen or extreme pH levels. Trend analysis identifies gradual changes, such as increasing turbidity or ammonia levels over time. Persistence analysis ensures that only sustained abnormal conditions trigger actions, reducing false alarms.

For instance, temporary fluctuations in turbidity caused by feeding do not trigger maintenance tasks. However, if turbidity remains high for an extended period, the system identifies it as a potential problem and generates a maintenance recommendation.

This multi-layered analysis approach improves accuracy and ensures that system outputs are meaningful and reliable.

6.5 Smart Maintenance Reminder Results

The Smart Maintenance Reminder module successfully converts sensor data into actionable maintenance tasks. The system generates reminders based on real-time conditions rather than fixed schedules.

Several scenarios were observed during testing. Sustained high turbidity levels triggered filter cleaning recommendations, while repeated drops in dissolved oxygen prompted suggestions to inspect aeration systems. Increased ammonia levels resulted in alerts to perform water exchange or cleaning.

Each maintenance task includes details such as type, priority, and timestamp, allowing users to respond effectively. The ability to acknowledge and track tasks through the dashboard enhances usability and accountability.

This demonstrates the effectiveness of Condition-Based Maintenance in reducing unnecessary work while ensuring timely intervention.

6.6 Alert System Results

The alert system performs efficiently in detecting and notifying abnormal conditions. Alerts are generated promptly when sensor values exceed safe limits or when trends indicate potential risks.

The response time for alert generation is typically within a few seconds, ensuring timely notification. Alerts are displayed on the dashboard and can also be delivered through additional channels such as SMS.

The cooldown mechanism prevents repeated alerts for the same condition, improving user experience. This ensures that notifications remain relevant and actionable.

The alert system significantly enhances the ability of farmers to respond quickly to environmental changes.

6.7 Environmental Event Impact Analysis

Environmental factors such as rainfall and temperature changes have a noticeable impact on water quality parameters. The system successfully captures these variations and responds accordingly.

For example, simulated rainfall conditions resulted in changes in pH and turbidity levels. The system detected these changes and generated appropriate alerts. This demonstrates the importance of event-based monitoring in aquaculture systems.

By incorporating environmental awareness, the system improves its ability to handle real-world scenarios and reduces the risk of sudden water quality deterioration.

6.8 System Performance Evaluation

The overall system performance is stable and reliable. The ESP32 microcontroller operates continuously without significant issues, and data transmission through MQTT remains consistent.

The backend system processes incoming data efficiently, and the dashboard updates in real time. Users can access system data with minimal delay, ensuring effective monitoring.

The system also demonstrates resilience to network interruptions. Data buffering mechanisms allow temporary storage of sensor data, which is synchronized once the connection is restored.

This confirms that the system is suitable for continuous operation in practical environments.

6.9 Usability and User Experience

The system provides a user-friendly interface that allows farmers to easily monitor water conditions and manage maintenance tasks. The dashboard presents information in a clear and organized manner, using visual elements such as charts and summaries.

Users can quickly understand system outputs, respond to alerts, and complete maintenance tasks with minimal effort. The simplicity of the interface ensures accessibility for users with different levels of technical knowledge.

The integration of monitoring, alerts, and maintenance features into a single platform improves overall usability and efficiency.

6.10 Discussion

The results demonstrate that the proposed system effectively addresses the limitations of traditional aquaculture monitoring methods. By combining real-time sensing with intelligent analysis, the system provides a comprehensive solution for water quality management.

One of the key strengths of the system is its ability to convert raw sensor data into meaningful actions. The integration of Condition-Based Maintenance ensures that maintenance tasks are performed based on actual conditions, reducing unnecessary effort and improving efficiency.

The inclusion of ammonia and CO₂ monitoring enhances the system's capability to detect toxic conditions, which are often overlooked in basic monitoring systems. This adds significant value to the overall solution.

The system also demonstrates strong performance in terms of reliability, scalability, and usability. However, certain limitations are observed, such as the need for regular sensor calibration and the influence of environmental factors on sensor readings.

Future improvements can focus on enhancing sensor accuracy, integrating predictive analytics, and expanding system capabilities.

6.11 Summary

This chapter presented the results and discussion of the Smart Maintenance Reminder and Water Quality Monitoring System. The findings confirm that the system provides accurate monitoring, effective analysis, and practical maintenance recommendations.

The integration of IoT technology with condition-based maintenance demonstrates a significant improvement over traditional method. The system offers a reliable and efficient solution for managing water quality in aquaculture environments.

7 CONCLUSION AND FUTURE WORK

7.1 Overview

This chapter presents the overall conclusion of the Smart Maintenance Reminder and Water Quality Monitoring System and summarizes the key outcomes achieved through the research. It also discusses the limitations encountered during the development process and outlines potential improvements for future work.

The objective of this chapter is to evaluate how effectively the proposed system addresses the identified problem and to highlight its contribution to smart aquaculture practices.

7.2 Conclusion

This study focused on designing and developing an IoT-based system to improve water quality monitoring and maintenance management in aquaculture environments. The proposed Smart Maintenance Reminder and Water Quality Monitoring System successfully integrates real-time sensing, data analysis, and intelligent decision-making into a single platform.

The system addresses the key limitations of traditional aquaculture practices, where monitoring is often manual and maintenance is performed based on fixed schedules. By

introducing continuous monitoring using IoT sensors, the system provides accurate and real-time insights into water quality conditions.

The implementation demonstrates that essential parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and CO₂ can be effectively monitored using low-cost sensor technologies. The integration of these sensors with the ESP32 microcontroller enables efficient data acquisition and transmission.

One of the most significant contributions of this study is the application of Condition-Based Maintenance in aquaculture. The system analyzes sensor data to generate maintenance tasks dynamically, ensuring that actions are performed only when required. This approach reduces unnecessary effort and helps prevent water quality deterioration.

The alert system further enhances the effectiveness of the solution by providing timely notifications for abnormal conditions. Farmers are able to respond quickly to changes in the environment, reducing the risk of fish stress and mortality.

The results confirm that the system is capable of providing accurate monitoring, reliable communication, and meaningful insights. The user-friendly dashboard allows easy interaction with the system, making it suitable for practical use in small and medium-scale fish farms.

Overall, the proposed system achieves its main objective of providing a low-cost, intelligent, and integrated solution for water quality management in aquaculture.

7.3 Achievements of the Study

The research successfully achieved several important outcomes. A complete IoT-based monitoring system was designed and implemented, integrating hardware, software, and cloud components into a unified solution.

Real-time monitoring of multiple water quality parameters was established using sensor technologies. The system demonstrated the ability to collect, process, and transmit data efficiently, ensuring continuous operation.

The development of the Smart Maintenance Reminder module introduced a practical implementation of Condition-Based Maintenance in aquaculture. This feature allows maintenance tasks to be generated based on actual system conditions rather than predefined schedules.

The alert system was successfully implemented, providing timely notifications for abnormal conditions and improving response time. The system also demonstrated reliable performance under continuous operation, confirming its suitability for real-world applications.

In addition, the inclusion of ammonia and CO₂ monitoring represents a novel contribution, as these parameters are often not included in low-cost monitoring systems.

7.4 Limitations of the Study

Although the system performs effectively, several limitations were identified during the development and testing phases.

Sensor accuracy depends on proper calibration, and environmental factors such as temperature and humidity can affect sensor readings. The ammonia sensor, in particular, may require further refinement to improve accuracy in different conditions.

The system currently relies on threshold-based and rule-based analysis for generating maintenance tasks. While this approach is effective, it does not incorporate advanced predictive models that could improve decision-making.

Another limitation is the dependency on network connectivity for real-time data transmission. Although the system includes basic handling for network interruptions, prolonged connectivity issues may affect real-time monitoring.

The implementation is also limited to a prototype designed for small-scale aquaculture environments. Scaling the system for large commercial farms would require additional modifications and optimization.

7.5 Future Work

Several enhancements can be made to improve the system in future developments. One of the key areas is the integration of advanced machine learning models for predictive analysis. This would allow the system to forecast water quality changes and recommend preventive actions more accurately.

The ammonia sensing mechanism can be further improved by developing more accurate and reliable sensors specifically designed for water-based environments. This would enhance the system's ability to detect toxic conditions.

The system can also be extended to support multi-pond monitoring, allowing users to manage multiple aquaculture units within a single platform. This would be particularly useful for larger farms.

Developing a dedicated mobile application would improve accessibility and user experience. Features such as push notifications and offline data access can further enhance usability.

Integration with automated control systems, such as aerators and water pumps, can also be considered. This would enable the system to not only monitor conditions but also take corrective actions automatically.

In addition, cloud-based analytics can be expanded to include advanced reporting and benchmarking features, helping farmers compare performance across different production cycles.

7.6 Final Remarks

This research demonstrates the potential of IoT technology in transforming traditional aquaculture practices into smart, data-driven systems. The Smart Maintenance Reminder and Water Quality Monitoring System provides a practical and affordable solution for improving water quality management and operational efficiency.

By combining real-time monitoring, intelligent analysis, and maintenance planning, the system supports better decision-making and reduces risks associated with poor water

conditions. The study highlights the importance of integrating multiple technologies to create a comprehensive solution that meets real-world needs.

The proposed system lays a strong foundation for future developments in smart aquaculture and contributes to the advancement of sustainable and efficient fish farming practices.

8 COMMERCIALIZATION AND ENTREPRENEURSHIP POTENTIAL

8.1 Overview

This chapter discusses the commercialization potential of the Smart Maintenance Reminder and Water Quality Monitoring System and its feasibility as a marketable product. It evaluates how the proposed system can be transformed from a research prototype into a practical solution for the aquaculture industry. The chapter also explores possible business models, target markets, and opportunities for entrepreneurship.

8.2 Market Need and Opportunity

Aquaculture is a rapidly growing industry worldwide, and in countries like Sri Lanka, small and medium-scale fish farming plays a significant role in food production and income generation. However, many farmers still rely on traditional methods for monitoring water quality and managing maintenance activities.

The lack of affordable and intelligent monitoring systems creates a strong market demand for low-cost technological solutions. Existing commercial systems are often expensive and designed for large-scale operations, making them inaccessible to small-scale farmers.

The proposed system addresses this gap by providing a cost-effective and easy-to-use solution that combines real-time monitoring with intelligent maintenance recommendations. This creates a significant opportunity for commercialization, particularly in developing regions where affordable solutions are needed.

8.3 Product Concept

The system can be developed into a complete product that includes both hardware and software components. The hardware unit consists of sensors and an ESP32-based controller that can be easily installed in fish ponds.

The software component includes a cloud-based backend and a user-friendly dashboard accessible through web or mobile applications. Together, these components provide a complete solution for monitoring water quality and managing maintenance tasks.

The product can be designed as a plug-and-play system, allowing farmers to install and use it with minimal technical knowledge. This simplicity is important for encouraging adoption among non-technical users.

8.4 Business Model

The commercialization of the system can follow a hybrid business model that combines hardware sales with subscription-based services.

The hardware component can be sold as a one-time purchase, including sensors, controller, and installation support. The software platform can be offered as a subscription service, providing access to real-time data, alerts, analytics, and maintenance features.

Different pricing tiers can be introduced based on features such as the number of ponds, level of analytics, and additional services like SMS alerts. This flexible pricing model allows the system to cater to different types of users.

8.5 Target Customers

The primary target customers for the system are small and medium-scale aquaculture farmers. These users require affordable and practical solutions to improve their operations.

Secondary customers may include aquaculture training institutes, research organizations, and government agencies involved in agricultural development. These organizations can use the system for monitoring, training, and demonstration purposes.

In addition, the system can be adapted for use in other water-based environments such as shrimp farming, hatcheries, and water treatment facilities, expanding its market potential.

8.6 Competitive Advantage

The proposed system offers several advantages compared to existing solutions. One of the main strengths is its low-cost design, which makes it accessible to small-scale farmers.

The integration of multiple sensors, including ammonia and CO₂ monitoring, provides a more comprehensive view of water quality than basic systems. The inclusion of the Smart Maintenance Reminder module adds further value by converting data into actionable insights. Unlike many existing systems that only provide monitoring, this system focuses on decision support, helping users take appropriate actions. The use of IoT and cloud technologies ensures scalability and remote accessibility.

These features create a strong competitive advantage in the aquaculture technology market.

8.7 Implementation and Deployment Strategy

To commercialize the system, a phased deployment strategy can be followed. Initially, the system can be introduced as a pilot project in selected farms to test performance and gather user feedback.

Based on the results, improvements can be made before launching the product on a larger scale. Partnerships with aquaculture suppliers, local distributors, and government agencies can help expand the reach of the product.

Training programs and user support services can also be provided to ensure successful adoption. Demonstrations and workshops can help farmers understand the benefits of the system.

8.8 Financial Feasibility

The system is designed to be cost-effective, making it suitable for commercialization. The use of affordable sensors and microcontrollers reduces the overall production cost.

Revenue can be generated through hardware sales and subscription fees for software services. Additional income can be obtained through maintenance services, upgrades, and support packages.

By reducing fish mortality and improving operational efficiency, the system provides economic benefits to users, making it a valuable investment.

8.9 Challenges and Risks

Despite its potential, the commercialization of the system may face certain challenges. One of the main challenges is the adoption of new technology by traditional farmers who may be hesitant to use digital systems.

Sensor reliability and maintenance may also be a concern, particularly in harsh environmental conditions. Ensuring consistent performance requires proper calibration and protection of hardware components.

Another challenge is network connectivity in rural areas, which may affect real-time data transmission. This can be addressed by incorporating offline capabilities and alternative communication methods.

Competition from existing products and rapid technological advancements may also pose risks. Continuous innovation and improvement are necessary to remain competitive.

8.10 Future Business Expansion

The system has strong potential for expansion into new markets and applications. Future developments can include integration with automated control systems, enabling the system to adjust aeration or water flow automatically.

Advanced analytics and artificial intelligence can be added to provide predictive insights and optimization recommendations. Mobile applications can be enhanced to improve accessibility and user engagement.

The system can also be expanded to support large-scale aquaculture operations, offering advanced features such as multi-pond management and centralized monitoring.

8.11 Summary

This chapter discussed the commercialization and entrepreneurship potential of the Smart Maintenance Reminder and Water Quality Monitoring System. The system addresses a clear market need and offers a practical, affordable, and scalable solution for aquaculture management.

The combination of hardware and software components, along with a flexible business model, creates strong opportunities for commercialization. While challenges exist, they can be addressed through proper planning and continuous improvement.

The proposed system has the potential to become a valuable product in the aquaculture industry, supporting the transition toward smart and sustainable farming practices.

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